

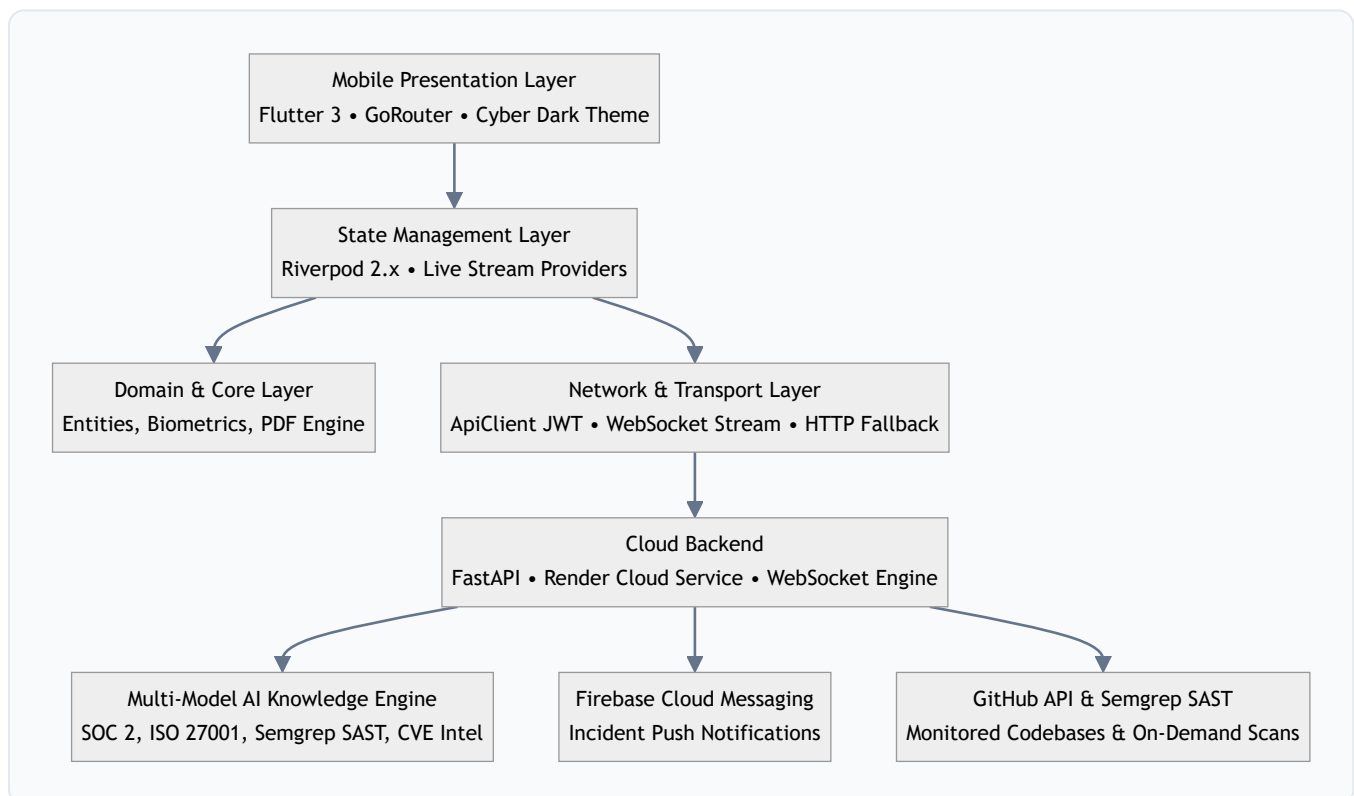
# SecurePulse

 FLUTTER 3.X DART 3.X FASTAPI CLOUD BACKEND FIREBASE FCM PUSH

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**SecurePulse** is a next-generation enterprise cybersecurity and SOC mobile operations platform built for security analysts, incident responders, and CISOs. Designed with a high-contrast dark cyberpunk aesthetic, hardware-backed biometric security, live WebSocket threat streaming, dynamic AI security assistance, real GitHub SAST repository scans, and one-tap executive PDF compliance report exports.

## Architecture & Tech Stack



## Core Technologies

- Mobile Frontend:** Flutter 3 (Dart 3), Riverpod State Management, GoRouter, `flutter_animate`, `fl_chart`.

- **Cloud Backend:** Python 3.11, FastAPI, Uvicorn, WebSockets, JWT Authentication, Render Cloud.
- **Push Notification Pipeline:** Google Firebase Cloud Messaging (FCM) & `firebase_admin` SDK.
- **Document Engine:** Pure vector `pdf` engine with native `share_plus` and `path_provider` export.
- **Hardware Security:** `local_auth` biometric authentication (Fingerprint / Face ID).

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## Key Platform Capabilities

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### 1. Executive Security Operations Dashboard

- Real-time **Security Posture Score** (94/100, Grade **A+**) with animated gauge rings and vulnerability charts.
- Live system status monitors for Cloud API, WebSockets, GitHub integration, and AI engine.
- Instant access to quick actions: One-tap SAST scans, SOC incident triage, AI chat, and compliance reports.

### 2. Real-Time WebSocket SOC Threat Streaming

- Subscribes to live WebSocket telemetry ( `wss://.../ws/alerts` ) for instant incident detection.
- Resilient architecture with **automatic HTTP long-polling fallback** if network proxies block raw sockets.
- Incident severity badges (Critical, High, Medium, Low), attack vector classifications, and IP origin telemetry.

### 3. GitHub Repository Integration & Live SAST Scans

- Monitored codebases include `NATTOMR/secureguard-mobile` and `NATTOMR/secureguard-backend` .
- One-tap "**Trigger Immediate SAST Scan**" dispatches real-time Semgrep & Secret scanning across cloud repositories.
- Live telemetry sync updates repository health grades and findings counts on your phone.

### 4. Dynamic AI Security Copilot ( `SecurePulse AI Agent` )

- Expert cybersecurity knowledge engine for **SOC 2 Type II**, **ISO/IEC 27001:2022**, and zero-day threat patches.
- Generates tailored remediation patches, `iptables` / `ufw` /Cloudflare firewall rules, and custom Semgrep YAML rules.

### 5. Executive PDF Compliance Report Export

- One-tap generation of official audit PDFs for:
- **SOC 2 Type II** (Trust Services Criteria CC6.1 - CC8.1)
- **ISO/IEC 27001:2022** (Annex A.5 - A.8)
- **PCI-DSS v4.0 & HIPAA Security Rule**

- Includes cryptographic SHA256 audit verification stamps and direct export via the native Android Share Sheet (Email, Slack, WhatsApp, Google Drive).

## 6. 🎨 Custom Cyber Shield Launcher & Splash Branding

- Custom dark obsidian metallic app icon with a **neon cyan holographic cyber defense shield and circuit lock core**.
- Android adaptive launcher icons ( `mdpi` , `hdpi` , `xhdpi` , `xxhdpi` , `xxxhdpi` ) and shimmering animated splash screen.

## 📁 Project Directory Structure

```
secureguard-mobile/
├── android/                # Android native configuration & Gradle scripts
│   ├── app/
│   │   ├── google-services.json # Firebase configuration
│   │   └── src/main/res/        # Custom adaptive cyber launcher icons
├── assets/
│   └── images/                # High-resolution app icon & branding assets
├── lib/
│   ├── app.dart              # Root MaterialApp & theme configuration
│   ├── main.dart             # App initialization (Firebase, Biometrics, Logging)
│   ├── core/                 # Core infrastructure layer
│   │   ├── config/           # Environment & backend URL configurations
│   │   ├── constants/        # AppColors, AppStrings, AppAssets
│   │   ├── network/          # ApiClient (JWT auto-auth), WebSocketService (with fallback)
│   │   ├── router/           # GoRouter with bottom navigation shell
│   │   ├── services/         # BiometricService, NotificationService, ReportPdfService
│   │   ├── storage/          # SecureStorageService (JWT token storage)
│   │   ├── theme/            # Cyber Dark theme tokens & Google Fonts
│   │   └── widgets/          # SGCard, SGButton, SGAppBar, SGStatisticCard, etc.
│   ├── features/             # Feature-First modular architecture
│   │   ├── ai/               # SecurePulse AI Agent chat screen & repository
│   │   ├── alerts/           # SOC incident alerts & incident detail investigation
│   │   ├── auth/             # Login screen with GitHub, Email, Biometric auth
│   │   ├── dashboard/        # Executive security dashboard & charts
│   │   ├── findings/         # Vulnerability findings & severity filters
│   │   ├── profile/          # Analyst profile & credentials
│   │   ├── reports/          # Executive compliance reports & PDF export
│   │   ├── repositories/     # Monitored codebases & SAST scan triggers
│   │   ├── scans/            # Scan history & detailed log viewer
│   │   ├── settings/         # Environment switching & platform settings
│   │   └── splash/           # Branded animated splash screen
│   └── providers/            # Riverpod providers & live stream notifiers
└── test/                     # Unit & integration test suite (15/15 passing)
```

## Complete Documentation Suite

Comprehensive technical, architectural, and operational documentation is available in the [docs/](#) directory:

| Document                                                                                                                   | Description                                                                     |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|  <a href="#">Project Technical Report</a> | Executive project dossier, problem matrix, test coverage, and security audit.   |
|  <a href="#">Architecture Mindmap</a>     | Visual Mermaid system architecture, dual mode dataflows, and navigation graphs. |
|  <a href="#">Operations Walkthrough</a>   | End-to-end screen workflows, triage steps, and developer user manual.           |
|  <a href="#">Product Roadmap</a>          | Strategic evolution phases, milestone timelines, and feature deliverables.      |
|  <a href="#">API Integration Guide</a>    | FastAPI endpoint contracts, WebSocket streaming, and deployment guides.         |

## Getting Started

### Prerequisites

- [Flutter SDK](#) ≥ 3.0.0
- [Dart SDK](#) ≥ 3.0.0
- Android Studio / Android SDK Platform 34+

### Setup & Run

```
# 1. Clone repository
git clone https://github.com/NATTOMR/secureguard-mobile.git
cd secureguard-mobile

# 2. Fetch dependencies
flutter pub get

# 3. Run test suite
flutter test

# 4. Run on connected device or emulator
flutter run
```

## Building Release APK

```
flutter build apk --release
```

The compiled APK will be generated at `build/app/outputs/flutter-apk/app-release.apk` (and mirrored at `securepulse-release.apk` in the root folder).

## Backend Cloud Endpoints

The mobile app connects to the live **SecurePulse Cloud Service** hosted on Render:

| Endpoint                                | Method | Description                              |
|-----------------------------------------|--------|------------------------------------------|
| <code>/v1/auth/login</code>             | POST   | Analyst JWT authentication               |
| <code>/v1/dashboard</code>              | GET    | Overall security posture score & metrics |
| <code>/v1/alerts</code>                 | GET    | Active SOC security incidents            |
| <code>/ws/alerts</code>                 | WSS    | Real-time WebSocket threat stream        |
| <code>/v1/repositories</code>           | GET    | Monitored codebases telemetry            |
| <code>/v1/repositories/{id}/scan</code> | POST   | Trigger on-demand SAST security scan     |
| <code>/v1/ai/chat</code>                | POST   | Live AI Security Copilot query           |
| <code>/v1/reports</code>                | GET    | Executive compliance report logs         |

## Security & Privacy

- **Zero-Trust Client Sessions:** Hardware-backed biometric authentication with auto-expiring JWT bearer tokens.
- **Encrypted Transmission:** All cloud traffic enforced over **TLS 1.3 / WSS** with strict CORS and rate limiting.
- **Push Protection:** Automated secret scanning to prevent credential leakage.

## License

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